

1. Slicing and Replacement

Given:

```
a = np.arange(1, 26).reshape(5, 5)
```

Replace all elements in the **middle 3×3 subarray** with 0.

3. Conditional Replacement

Given:

```
a = np.array([12, 5, 18, 7, 25, 3, 14, 9, 30])
```

Replace values less than 10 with -1, values greater than 20 with 1, and leave everything else unchanged.

4. Row-wise Maximum

Given:

```
a = np.array([
    [12, 45, 23, 17],
    [56, 21, 34, 10],
    [18, 39, 27, 44]
])
```

1. Find the maximum of each row.
2. Find the index of the maximum element in each row.

5. Column-wise Operation

Given:

```
a = np.array([
    [2, 4, 6, 8],
    [1, 3, 5, 7],
    [10, 20, 30, 40]
])
```

1. Find the minimum of each column.
2. Find the index of the minimum element in each column.
3. Subtract the **column mean** from every element in that column.